

Section 6: Interdependencies Among Principles

1 Interdependencies Among Principles

The framework for addressing global regularity of the Navier–Stokes equations integrates four foundational principles: the Scale-Barrier Principle, Monotone Functional, Non-Sobolev Compactness, and Curvature-Based Regularity. These principles work synergistically, with each reinforcing the others to establish smoothness, boundedness, and global regularity.

1.1 Mapping the Interdependencies

To illustrate the interconnectedness of these principles, we analyze their interactions:

1. Scale-Barrier Principle: High-Frequency Control The Scale-Barrier Principle ensures that the energy cascade to high wave numbers is suppressed. By imposing spectral decay bounds:

$$E(k, t) \leq Ck^{-\beta}, \quad \beta > 3, \quad (1)$$

it prevents unbounded growth of velocity gradients and contributes to compactness. This principle provides:

- ****Input to Non-Sobolev Compactness:**** By ensuring finite energy at high wave numbers, it guarantees that solution sequences remain smooth in the generalized space H .
- ****Support for Curvature Constraints:**** High-frequency damping aligns with the curvature-based regularity framework, controlling the nonlinear term $C|\nabla u|^3$.

2. Monotone Functional: Global Energy Bounds The monotone functional $Q(t)$:

$$Q(t) = \|u(t)\|_{L^2}^2 + \|\nabla u(t)\|_{L^2}^2, \quad (2)$$

provides a tool for tracking global energy and enstrophy over time. Its evolution:

$$\frac{dQ}{dt} = -\nu\|\Delta u\|_{L^2}^2 + C\|\nabla u\|_{L^2}^3, \quad (3)$$

ensures that energy dissipation dominates nonlinear growth.

- ****Input to Scale-Barrier Principle:**** The boundedness of $Q(t)$ reinforces the spectral decay assumption by preventing infinite energy transfer to high wave numbers.
- ****Synergy with Non-Sobolev Compactness:**** Bounded global energy ensures that sequences of solutions in L^2 converge strongly.

3. Non-Sobolev Compactness: Smooth Solution Sequences Compactness in the non-Sobolev space H :

$$\|u\|_H = \|u\|_{L^2} + \|\nabla u\|_{L^2} + \sup_{k > k_0} \frac{E(k)}{k^\beta}, \quad \beta > 0, \quad (4)$$

ensures strong convergence of approximate solutions without relying on classical Sobolev embeddings. By leveraging spectral decay from the Scale-Barrier Principle and bounded energy from the Monotone Functional, it:

- ****Ensures Regularity:**** Smooth convergence in L^2 supports the transition from approximate solutions to smooth limits.
- ****Integrates with Curvature-Based Regularity:**** Compactness aids in proving the boundedness of velocity gradients, reinforcing curvature constraints.

4. Curvature-Based Regularity: Gradient Control The curvature evolution equation:

$$\frac{\partial}{\partial t} |\nabla u|^2 \leq -\text{Ric}(g) |\nabla u|^2 + C |\nabla u|^3, \quad (5)$$

highlights the role of Ricci curvature in damping velocity gradients. This principle:

- ****Utilizes Compactness:**** Smooth convergence ensures that velocity gradients remain bounded under curvature constraints.
- ****Reinforces the Scale-Barrier Principle:**** Gradient damping aligns with high-frequency energy suppression, ensuring global regularity.

1.2 Sub-Examples of Interactions

Example 3: Viscous Effects Across Principles Viscosity, represented by the term $\nu \Delta u$, plays a central role in reinforcing all four principles:

- ****Spectral Decay in the Scale-Barrier Principle:**** Viscosity ensures dissipation dominates nonlinear energy transfer at high wave numbers. By suppressing energy cascades beyond a critical threshold k_{crit} , viscosity directly enforces the spectral decay:

$$E(k, t) \leq C k^{-\beta}, \quad \beta > 3. \quad (6)$$

This decay prevents the formation of singularities by restricting energy accumulation in small scales.

- ****Dissipation Dominance in the Monotone Functional:**** In the energy evolution equation:

$$\frac{dQ}{dt} = -\nu \|\Delta u\|_{L^2}^2 + C \|\nabla u\|_{L^2}^3, \quad (7)$$

the viscous dissipation term $-\nu \|\Delta u\|_{L^2}^2$ counterbalances nonlinear growth $C \|\nabla u\|_{L^2}^3$. This ensures that $Q(t)$ remains globally bounded over time, preventing blowup.

- ****Strong Convergence in Non-Sobolev Compactness:**** The viscous term enhances compactness by dissipating high-frequency components of the solution. Specifically, the spectral control provided by $\nu\Delta u$ ensures:

$$\sup_{k>k_0} \frac{E(k)}{k^\beta} \rightarrow 0 \quad \text{as } k_0 \rightarrow \infty, \quad (8)$$

which is a key requirement for smooth L^2 convergence in the generalized space H .

- ****Gradient Damping in Curvature-Based Regularity:**** In the curvature evolution equation:

$$\frac{\partial}{\partial t} |\nabla u|^2 \leq -\text{Ric}(g) |\nabla u|^2 + \nu \Delta_g |\nabla u|^2 + C |\nabla u|^3, \quad (9)$$

viscosity ($\nu \Delta_g |\nabla u|^2$) synergizes with positive Ricci curvature $\text{Ric}(g)$ to dampen velocity gradients. This combined effect ensures that $|\nabla u|^2$ remains bounded, reinforcing global regularity.

References

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